CHE 201 Fall 2019 HW 1

Hussein Hussein

TOTAL POINTS

86 / 100

QUESTION 1

1 1 **10 / 10**

✓ **- 0 pts** Correct

- 2 pts Missing 1.9 and 1.10

- 0.5 pts The correct term for 1.3 is "control volume"

- 2 pts Control volume is B and Process is D

- 1 pts The correct answer for 1.10 is C (not A)

- 1 pts The correct answer for 1.3 (control volume) is B

- 2 pts The answer for 1.4 is E and 1.5 is H

QUESTION 2

2 2 **8 / 8**

✓ **- 0 pts** Correct

- 1 pts F) 0.54 ft3/s

- 1 pts C) 99.57 ft.lbf/s

- 1 pts E) 44 lbf/in2

- 0.5 pts G) 45.55 ft/s

- 0.5 pts H) 1 ton

- 0.5 pts A) 61.024 in3

- 0.5 pts D) 50 lb/min

- 0.5 pts B) 0.616 BTU

- 3 pts No calculation shown

- 0.5 pts C) 99.57 ft.lbf/s

- 1 pts H) 1 ton

- 1 pts Missing F) 0.54 ft3/s

- 1 pts No calculation shown for (H)

- 2 pts No units used for conversion factors in (D)-(G)

- 1 pts F) 0.54 ft3/s

- 0.5 pts Insufficient conversion factor used in (A)

- 1 pts No calculation shown for (A)

- 1 pts Overall use of secondary conversion factors

- 0.5 pts F) 0.54 ft3/s

- 1 pts G) 45.55 ft/s

- 0.5 pts E) wrong conversion factor

- 1 pts Missing the unit of the conversion factors

- 0.5 pts Use of secondary conversion factor in (G) and (F)

- 0.5 pts Use of secondary conversion factor in (D) and (F)

- 1 pts D) 50 lb/min

QUESTION 3

3 3 **5 / 5**

✓ **- 0 pts** Correct

- 1 pts a) 5.58x10^-4 kg

- 0.5 pts a) 1.993 x 10^-5 kmol

- 0.5 pts b) 50.175 m3/kmol

- 1 pts b) 1.792 m3/kg

- 0.5 pts a) 5.58x10^-4 kg

- 0.5 pts b) 1.792 m3/kg

- 1 pts b) 50.175 m3/kmol

- 1 pts a) 1.993 x 10^-5 kmol

QUESTION 4

4 4 **5 / 5**

✓ **- 0 pts** Correct

- 0.5 pts B) 277.8 gmoles

- 1.5 pts C) incorrect calculation

- 1 pts V = 1.08 m3

- 1 pts c) n = 1.67 x 10^26 molecules

- 2 pts B) 277.8 gmoles

QUESTION 5

5 5 **9 / 10**

- 0 pts correct

✓ **- 1 pts** Radius of piston was incorrect

- 10 pts no answer

- 5 pts Force balance incorrect

- 3 pts Force calculation incorrect

- **3 pts** 491N?

QUESTION 6

6 6 **8 / 8**

✓ **- 0 pts** Correct

- 2 pts Process 4 incorrect

- 8 pts Process In series on P-V

- 8 pts no answer

- 6 pts Pressure increase for process 2 not decrease

- 0 pts Click here to replace this description.

QUESTION 7

7 7 **8 / 8**

✓ **- 0 pts** Correct

- 8 pts no answer

- 4 pts 7b ?

- 2 pts unit calculation incorrect 7b

- 8 pts 1. unit incorrect or not show 2. calculation incorrect 7b

- 6 pts unit incorrect for both a and b

- 1 pts no unit for answer 7a

QUESTION 8

8 8 **10 / 10**

✓ **- 0 pts** Correct

- 10 pts which pressure? calculation?

- 2 pts unit wrong for kPa

- 5 pts pressure was wrong

- 10 pts no answer

- 8 pts which pressure

- 10 pts please provide the clear copy, cant read this

QUESTION 9

9 9 **8 / 8**

✓ **- 0 pts** Correct

- 4 pts Didn't use (rho).g.h

- 2 pts Wrong conversion factor (should convert from mm to m and not mm3 to m3)

- 0.5 pts Missing the pressure unit (Pa)

- 1 pts Didn't convert from Pa to kPa in the end. Should have made 101321 Pa = 101.3 kPa

- 8 pts Missing answer

- 5 pts Illegible answer

- 1 pts 13,590 x 9.81 x 0.76 = 101,321.6 (not 101,218.32)

- 2 pts Just made a simple unit conversion from mmHg to kPa (didn't prove the relation).

QUESTION 10

10 10 **8 / 8**

✓ **- 0 pts** Correct

- 5 pts Didn't calculate pressure at the exit

- 8 pts Missing answer

- 3 pts The compression ratio should be used as the ratio of the absolute pressure of the gas at the exit to the absolute inlet pressure

- 2 pts Hard to read answer

- 3 pts Incorrect conversion at the end. Answer is 160 psia.

QUESTION 11

11 11 **7 / 8**

- 0 pts Correct

- 1.5 pts Wrong conversion factor from bar to Pa; 1 bar = 10^5 Pa.

- 2 pts (rho).g.h = 10^3x9.81x30 = 2.94x10^5 Pa

✓ **- 1 pts** **Didn't show the conversion from Pa to bar**

- 1 pts Wrong unit in the end (should be Pa and not N)

- 1 pts Wrong calculation for (rho).g.h. The right answer is 2.94x10^5 Pa

- 1.5 pts Final answer should be P-air = 1.21x10^5 Pa = 1.21 bar

- 1 pts Didn't convert from Pa to bar at the end

- 4 pts Did not finish the answer

- 8 pts Missing Answer

- 1 pts Forgot to carry the 10^5 term in the end

QUESTION 12

12 12 **0 / 12**

- 0 pts Correct

✓ **- 12 pts** **Missing answer**

- **2.5 pts** Missing analysis of the environmental considerations for coal power plants
- **2.5 pts** Missing analysis of the environmental considerations for natural gas power plants
- **2.5 pts** Missing analysis of the environmental considerations for nuclear gas power plants
- **2.5 pts** Missing analysis of the environmental considerations for for hydroelectric power plants
- **1 pts** Doesn't have at least 3 references
- **2 pts** Did not present references
- **1 pts** Missing the percentage distribution for all four sources
- **2.5 pts** Missing how the environmental aspects impact plant design, operation and cost
- **1 pts** Unclear definition of the fossil fuels being addressed
- **1.5 pts** It's not clearly stated how environmental aspects impact plant design, operation and cost for all the sources of power
- **1.5 pts** Did not consider particularities of coal and natural gas, but fossil fuels in general
- **5 pts** Did not determine at least three significant environmental considerations associated with each of the four sources of power.
- **0.5 pts** Listed only 2 environmental considerations for hydroelectric power plants
- **3 pts** Did not list at least 3 environmental considerations and their impacts for both coal and natural power plants, each.
- **1 pts** Did not list at least 3 environmental considerations for nuclear power plants and how they impact plant design, operation
- **2.5 pts** It's not clearly stated how environmental aspects impact plant design, operation and cost for coal, nuclear and hydroelectric power plants
- **0.5 pts** Missing the percentage for natural gas
- **1 pts** Did not list at least 3 environmental considerations for natural gas power plants
- **1 pts** Missing how the environmental aspects impact on coal power plant design, operation and cost
- **1.5 pts** It's not clearly stated the impact of all environmental considerations mentioned for nuclear, natural gas and hydroelectric power plants
- **3 pts** Missing the analysis of how the environmental aspects impact plant design, operation and cost for the power plants
- **1 pts** It's not clearly stated the impact of all environmental considerations mentioned for natural gas and hydroelectric power plants design, operation and costs
- **1 pts** Missing how the environmental aspects impact on nuclear power plant design, operation and cost
- **0.5 pts** Missing a list of the references used
- **0.5 pts** Missing the percentage for coal
- **2 pts** It's not clearly stated the impact of all environmental considerations mentioned for all four sources of power plants
- **0.5 pts** Did not separate the percentage of fossil fuel energy into natural gas and coal
- **2.5 pts** Did not list at least 3 (three) environmental aspects and their impact plant design, operation and cost for nuclear and hydroelectric power plants
- **0.5 pts** Listed only 2 environmental considerations for natural gas power plants
- **0.5 pts** Listed only 2 environmental considerations for coal power plants
- **1.5 pts** Listed only 1 environmental aspect for hydroelectric power plant
- **1.5 pts** It's not clearly stated how those environmental aspects impact plant design, operation and cost for coal and natural gas power plants
- **1.5 pts** It's not clearly stated how those environmental aspects impact plant design, operation and cost for nuclear and hydroelectric power plants

HW #1 due 9/6

Hussein Hussein

1. I
2.) G
3.) B
4.) E
5. H
6.) D
7.) J
8.) A
9.) F
10.) C

a.) $1\cancel{L} \cdot \frac{0.0353\cancel{ft^3}}{1\cancel{L}} \cdot \left(\frac{12\,in}{1\cancel{ft}}\right)^3 = \boxed{61\,in^3}$

b.) $650\cancel{J} \cdot \frac{1\,\cancel{kJ}}{10^3\cancel{J}} \cdot \frac{1\,Btu}{1.0551\,\cancel{kJ}} = \boxed{0.616\,Btu}$

C.) $0.135\,kW \cdot \frac{3413\,Btu/h}{1\,kW} \cdot \frac{1\,h}{3600\,s} \cdot \frac{778.17}{1\,btu} = \boxed{99.596\,\frac{ft \cdot lb}{s}}$

D. $378\,g/s \cdot \frac{1\,kg}{10^3\,g} \cdot \frac{1\,lb}{0.4536\,kg} \cdot \frac{60\,s}{1\,mi} = \boxed{50\,\frac{lb}{min}}$

E.) $304\,kPa \cdot \frac{1\,lbf/in^3}{6894.8\,Pa} \cdot \frac{10^3\,Pa}{1\,kPa} = \boxed{44.09\,\frac{lbf}{in^2}}$

f.) $55\,\frac{m^3}{h} \cdot \frac{3.2808\,ft}{1\,m} \cdot \frac{1\,h}{3600\,s} = \boxed{0.54\,\frac{ft^3}{s}}$

g.) $50\,\frac{km}{h} \cdot \frac{10^3\,m}{1\,km} \cdot \frac{3.2808\,ft}{1\,m} \cdot \frac{1}{3600\,s} = \boxed{45.57\,\frac{ft}{s}}$

1 1 **10 / 10**

- ✓ **- 0 pts** Correct
- **- 2 pts** Missing 1.9 and 1.10
- **- 0.5 pts** The correct term for 1.3 is "control volume"
- **- 2 pts** Control volume is B and Process is D
- **- 1 pts** The correct answer for 1.10 is C (not A)
- **- 1 pts** The correct answer for 1.3 (control volume) is B
- **- 2 pts** The answer for 1.4 is E and 1.5 is H

HW #1 due 9/6

Hussein Hussein

1, I

2.) G

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9.) F

10.) C

2.) $1\,\cancel{L} \cdot \frac{0.0353\,\cancel{ft}^3}{1\,\cancel{L}} \cdot \left(\frac{12\,in}{1\,\cancel{ft}}\right)^3 = \boxed{61\,in^3}$

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D. $378\,g/s \cdot \frac{1\,kg}{10^3\,g} \cdot \frac{1\,lb}{0.45361\,kg} \cdot \frac{60\,s}{1\,mi} = \boxed{50\,\frac{lb}{min}}$

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f.) $55\,\frac{m^3}{h} \cdot \left(\frac{3.2808\,ft}{1\,m}\right)^3 \cdot \frac{1\,h}{3600\,s} = \boxed{0.54\,\frac{ft^3}{s}}$

g.) $50\,\frac{km}{h} \cdot \frac{10^3\,m}{1\,km} \cdot \frac{3.2808\,ft}{1\,m} \cdot \frac{1}{3600\,s} = \boxed{45.57\,\frac{ft}{s}}$

5 *5

5 *

h.) $8896 N \cdot \frac{1 lbf}{4.4482 N} \cdot \frac{1 ton}{2000 lbf} = \boxed{1 \text{ ton}}$

3.) a.) $\frac{1.2 \cdot 10^{22} \text{ m of } N_2}{6.022 \cdot 10^{23} \text{ mol/mole}} = \frac{0.0199 \text{ mole}}{10^3 \text{ mole}} = \boxed{0.000019927 \text{ kmol}}$

Nitrogen gas = N_2 = $28.02g \cdot 1.9927 \cdot 10^{-5} = \boxed{5.58353 \cdot 10^{-4} \text{ kg}}$

b.) $\frac{10^{-3}}{5.58353 \cdot 10^{-4}} = 1.79 \frac{m^3}{kg}$

$\frac{10^{-3}}{1.9927 \cdot 10^{-5}} = \boxed{50.18 \frac{m^3}{kmol}}$

4.) A.) $V = m v = 5 \cdot 0.2160 = 1.08 m^3$

B.) $\frac{5}{18.02} \cdot \frac{1000}{1} = \boxed{277.5 \text{ moles}}$

C.) $6.022 \cdot 10^{23} \cdot 277.5 = \boxed{1.671 \cdot 10^{26} \text{ molecules}}$

2 2 **8 / 8**

✓ **- 0 pts** Correct

- 1 pts F) 0.54 ft3/s

- 1 pts C) 99.57 ft.lbf/s

- 1 pts E) 44 lbf/in2

- 0.5 pts G) 45.55 ft/s

- 0.5 pts H) 1 ton

- 0.5 pts A) 61.024 in3

- 0.5 pts D) 50 lb/min

- 0.5 pts B) 0.616 BTU

- 3 pts No calculation shown

- 0.5 pts C) 99.57 ft.lbf/s

- 1 pts H) 1 ton

- 1 pts Missing F) 0.54 ft3/s

- 1 pts No calculation shown for (H)

- 2 pts No units used for conversion factors in (D)-(G)

- 1 pts F) 0.54 ft3/s

- 0.5 pts Insufficient conversion factor used in (A)

- 1 pts No calculation shown for (A)

- 1 pts Overall use of secondary conversion factors

- 0.5 pts F) 0.54 ft3/s

- 1 pts G) 45.55 ft/s

- 0.5 pts E) wrong conversion factor

- 1 pts Missing the unit of the conversion factors

- 0.5 pts Use of secondary conversion factor in (G) and (F)

- 0.5 pts Use of secondary conversion factor in (D) and (F)

- 1 pts D) 50 lb/min

5*5

5*

h.) $8896N \cdot \frac{1 lbf}{4.4482N} \cdot \frac{1 ton}{2000 lbf} = \boxed{1 \text{ ton}}$

3.) a.) $\frac{1.2 \cdot 10^{22} \text{ m of } N_2}{6.022 \cdot 10^{23} \text{ mol/mole}} = \frac{0.0199 \text{ mole}}{10^3 \text{ mole}} = \boxed{0.0000199 27 \text{ kmol}}$

Nitrogen gas $= N_2 = 28.02g \cdot 1.9927 \cdot 10^{-5} = \boxed{5.58353 \cdot 10^{-4} \text{ kg}}$

b.) $\frac{10^{-3}}{5.58353 \cdot 10^{-4}} = 1.79 \frac{m^3}{kg}$

$\frac{10^{-3}}{1.9927 \cdot 10^{-5}} = \boxed{50.18 \frac{m^3}{kmol}}$

4.) A.) $V = m v = 5 \cdot 0.2160 = 1.08 m^3$

B.) $\frac{5}{18.02} \cdot \frac{1000}{1} = \boxed{277.5 \text{ moles}}$

C.) $6.022 \cdot 10^{23} \cdot 277.5 = \boxed{1.671 \cdot 10^{26} \text{ molecules}}$

3 3 **5 / 5**

- ✓ **- 0 pts** Correct
- **- 1 pts** a) 5.58x10^-4 kg
- **- 0.5 pts** a) 1.993 x 10^-5 kmol
- **- 0.5 pts** b) 50.175 m3/kmol
- **- 1 pts** b) 1.792 m3/kg
- **- 0.5 pts** a) 5.58x10^-4 kg
- **- 0.5 pts** b) 1.792 m3/kg
- **- 1 pts** b) 50.175 m3/kmol
- **- 1 pts** a) 1.993 x 10^-5 kmol

5 *5

5 *

h.) $8896N \cdot \frac{1 lbf}{4.4482 N} \cdot \frac{1 ton}{2000 lbf} = \boxed{1 \text{ ton}}$

3.) a.) $\frac{1.2 \cdot 10^{22} \text{ m of } N_2}{6.022 \cdot 10^{23} \text{ mol/mole}} = \frac{0.0199 \text{ mole}}{10^3 \text{ mole}} = \boxed{0.0000199927 \text{ kmol}}$

Nitrogen gas = N_2 = $28.02g \cdot 1.9927 \cdot 10^{-5} = \boxed{5.58353 \cdot 10^{-4} \text{ kg}}$

b.) $\frac{10^{-3}}{5.58353 \cdot 10^{-4}} = 1.79 \frac{m^3}{kg}$

$\frac{10^{-3}}{1.9927 \cdot 10^{-5}} = \boxed{50.18 \frac{m^3}{kmol}}$

4.) A.) $V = m v = 5 \cdot 0.2160 = 1.08 m^3$

B.) $\frac{5}{18.02} \cdot \frac{1000}{1} = \boxed{277.5 \text{ moles}}$

C.) $6.022 \cdot 10^{23} \cdot 277.5 = \boxed{1.671 \cdot 10^{26} \text{ molecules}}$

4 4 **5 / 5**

- ✓ **- 0 pts** Correct
- **- 0.5 pts** B) 277.8 gmoles
- **- 1.5 pts** C) incorrect calculation
- **- 1 pts** V = 1.08 m3
- **- 1 pts** c) n = 1.67 x 10^26 molecules
- **- 2 pts** B) 277.8 gmoles

8.

5.) $\frac{\pi \cdot D^2}{4} = \frac{\pi \cdot 10cm}{4} = 78.54cm^2$

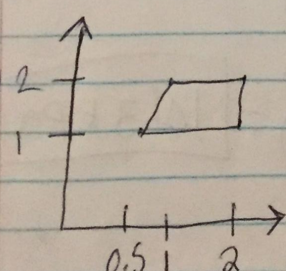
$3 \cdot 10^5 \frac{N}{m^2} \cdot 78.54 \cdot \left(\frac{1}{10}\right)^2 = 2356.2\,N$

$(78.54 - 0.8) \cdot 10^5 \cdot \left(\frac{1}{10}\right)^2 = 777.4\,N$

$25 \cdot 9.81 = 245.3\,N$

$F = 2356.2\,N - 245.3 - 777.4 = \boxed{1333.5\,N}$

6.)



5 5 **9 / 10**

- **- 0 pts** correct
- ✓ **- 1 pts Radius of piston was incorrect**
- **- 10 pts** no answer
- **- 5 pts** Force balance incorrect
- **- 3 pts** Force calculation incorrect
- **- 3 pts** 491N?

B.

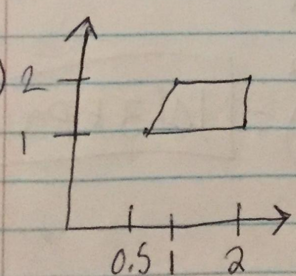
5.) $\frac{\pi \cdot D^2}{4} = \frac{\pi \cdot 10cm}{4} = 78.54 cm^3$

$3 \cdot 10^5 \frac{N}{m^2} \cdot 78.54 \cdot \left(\frac{1}{10}\right)^2 = 2356.2\,N$

$(78.54 - 0.8) \cdot 10^5 \cdot \left(\frac{1}{10}\right)^2 = 777.4\,N$

$25 \cdot 9.81 = 245.3\,N$

$f = 2356.2\,N - 245.3 - 777.4 = \boxed{1333.5N}$

6.) 

6 6 **8 / 8**

- ✓ **- 0 pts** Correct
- **- 2 pts** Process 4 incorrect
- **- 8 pts** Process In series on P-V
- **- 8 pts** no answer
- **- 6 pts** Pressure increase for process 2 not decrease
- **- 0 pts** Click here to replace this description.

7.) $P_{gas} = P_{atm} + \rho g h$

a.) $$\frac{(1.5-1)\cdot 10^5}{997 \cdot 9.8} = \boxed{5.11m}$$

b.) $$\frac{(1.3-1)\cdot 10^5 \ N/m^2}{13.59 \frac{g}{cm^3} \cdot \frac{1}{10^3} \cdot \frac{10^2}{1} \cdot 9.81} = \boxed{0.225m}$$

8.) pressure in the tank > ~~Press~~ atmospheric pressure therefore its gage pressure

$200 - 100 = \boxed{100 \ kP}$

9.) $13590 \cdot 9.8 \cdot 760 \cdot \left(\frac{1}{1000}\right)^2 = \boxed{101.3 \ kPa}$

10.) $5.5 + 14.5 = 20$ psia

$20 \cdot 8 = \boxed{160 \ psia}$

7 7 **8 / 8**

✓ **- 0 pts** **Correct**

- 8 pts no answer

- 4 pts 7b ?

- 2 pts unit calculation incorrect 7b

- 8 pts 1. unit incorrect or not show 2. calculation incorrect 7b

- 6 pts unit incorrect for both a and b

- 1 pts no unit for answer 7a

7.) $P_{gas} = P_{atm} + \rho g h$

a.) $\frac{(1.5-1)\cdot 10^5}{997 \cdot 9.8} = \boxed{5.11m}$

b.) $\frac{(1.3-1)\cdot 10^5 \ N/m^2}{13.59 \frac{g}{cm^3} \cdot \frac{1}{10^3} \cdot \frac{10^2}{1} \cdot 9.81} = \boxed{0.225m}$

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9.) $13590 \cdot 9.8 \cdot 760 \cdot \left(\frac{1}{1000}\right)^2 = \boxed{101.3 \ kPa}$

10.) $5.5 + 14.5 = 20$ psia

$20 \cdot 8 = \boxed{160 \text{ psia}}$

8 8 10 / 10

✓ **- 0 pts** Correct

- 10 pts which pressure? calculation?

- 2 pts unit wrong for kPa

- 5 pts pressure was wrong

- 10 pts no answer

- 8 pts which pressure

- 10 pts please provide the clear copy, cant read this

7.) $P_{gas} = P_{atm} + \rho g h$

a.) $\frac{(1.5-1)\cdot 10^5}{997 \cdot 9.8} = \boxed{5.11m}$

b.) $\frac{(1.3-1)\cdot 10^5 \ N/m^2}{13.59 \frac{g}{cm^3} \cdot \frac{1}{10^3} \cdot \frac{10^2}{1} \cdot 9.81} = \boxed{0.225m}$

8.) pressure in the tank > ~~Press~~ atmospheric pressure therefore its gage pressure

$200 - 100 = \boxed{100 \ kP}$

9.) $13590 \cdot 9.8 \cdot 760 \cdot \left(\frac{1}{1000}\right)^2 = \boxed{101.3 \ kPa}$

10.) $5.5 + 14.5 = 20$ psia

$20 \cdot 8 = \boxed{160 \text{ psia}}$

9 **9** **8 / 8**

- ✓ **- 0 pts** Correct
- **- 4 pts** Didn't use (rho).g.h
- **- 2 pts** Wrong conversion factor (should convert from mm to m and not mm3 to m3)
- **- 0.5 pts** Missing the pressure unit (Pa)
- **- 1 pts** Didn't convert from Pa to kPa in the end. Should have made 101321 Pa = 101.3 kPa
- **- 8 pts** Missing answer
- **- 5 pts** Illegible answer
- **- 1 pts** 13,590 x 9.81 x 0.76 = 101,321.6 (not 101,218.32)
- **- 2 pts** Just made a simple unit conversion from mmHg to kPa (didn't prove the relation).

7.) $P_{gas} = P_{atm} + \rho g h$

a.) $\dfrac{(1.5-1)\cdot 10^5}{997 \cdot 9.8} = \boxed{5.11m}$

b.) $\dfrac{(1.3-1)\cdot 10^5 \ N/m^2}{13.59 \frac{g}{cm^3} \cdot \frac{1}{10^3} \cdot \frac{10^2}{1} \cdot 9.81} = \boxed{0.225m}$

8.) pressure in the tank > ~~Press~~ atmospheric pressure therefore its gage pressure

$200 - 100 = \boxed{100 \ kP}$

9.) $13590 \cdot 9.8 \cdot 760 \cdot \left(\frac{1}{1000}\right)^2 = \boxed{101.3 \ kPa}$

10.) $5.5 + 14.5 = 20$ psia

$20 \cdot 8 = \boxed{160 \text{ psia}}$

10 10 **8 / 8**

✓ **- 0 pts** Correct

- 5 pts Didn't calculate pressure at the exit

- 8 pts Missing answer

- 3 pts The compression ratio should be used as the ratio of the absolute pressure of the gas at the exit to the absolute inlet pressure

- 2 pts Hard to read answer

- 3 pts Incorrect conversion at the end. Answer is 160 psia.

11.) $p = p + \rho g L$

$p = 4.15 - 9{,}8 \cdot 10^3 \cdot 30 = \boxed{1.21 \text{ bar}}$

11 11 7 / 8

- **- 0 pts** Correct
- **- 1.5 pts** Wrong conversion factor from bar to Pa; 1 bar = 10^5 Pa.
- **- 2 pts** (rho).g.h = 10^3x9.81x30 = 2.94x10^5 Pa
- ✓ **- 1 pts Didn't show the conversion from Pa to bar**
- **- 1 pts** Wrong unit in the end (should be Pa and not N)
- **- 1 pts** Wrong calculation for (rho).g.h. The right answer is 2.94x10^5 Pa
- **- 1.5 pts** Final answer should be P-air = 1.21x10^5 Pa = 1.21 bar
- **- 1 pts** Didn't convert from Pa to bar at the end
- **- 4 pts** Did not finish the answer
- **- 8 pts** Missing Answer
- **- 1 pts** Forgot to carry the 10^5 term in the end

12 12 **0 / 12**

- **- 0 pts** Correct
- ✓ **- 12 pts Missing answer**
- **- 2.5 pts** Missing analysis of the environmental considerations for coal power plants
- **- 2.5 pts** Missing analysis of the environmental considerations for natural gas power plants
- **- 2.5 pts** Missing analysis of the environmental considerations for nuclear gas power plants
- **- 2.5 pts** Missing analysis of the environmental considerations for for hydroelectric power plants
- **- 1 pts** Doesn't have at least 3 references
- **- 2 pts** Did not present references
- **- 1 pts** Missing the percentage distribution for all four sources
- **- 2.5 pts** Missing how the environmental aspects impact plant design, operation and cost
- **- 1 pts** Unclear definition of the fossil fuels being addressed
- **- 1.5 pts** It's not clearly stated how environmental aspects impact plant design, operation and cost for all the sources of power
- **- 1.5 pts** Did not consider particularities of coal and natural gas, but fossil fuels in general
- **- 5 pts** Did not determine at least three significant environmental considerations associated with each of the four sources of power.
- **- 0.5 pts** Listed only 2 environmental considerations for hydroelectric power plants
- **- 3 pts** Did not list at least 3 environmental considerations and their impacts for both coal and natural power plants, each.
- **- 1 pts** Did not list at least 3 environmental considerations for nuclear power plants and how they impact plant design, operation
- **- 2.5 pts** It's not clearly stated how environmental aspects impact plant design, operation and cost for coal, nuclear and hydroelectric power plants
- **- 0.5 pts** Missing the percentage for natural gas
- **- 1 pts** Did not list at least 3 environmental considerations for natural gas power plants
- **- 1 pts** Missing how the environmental aspects impact on coal power plant design, operation and cost
- **- 1.5 pts** It's not clearly stated the impact of all environmental considerations mentioned for nuclear, natural gas and hydroelectric power plants
- **- 3 pts** Missing the analysis of how the environmental aspects impact plant design, operation and cost for the power plants
- **- 1 pts** It's not clearly stated the impact of all environmental considerations mentioned for natural gas and hydroelectric power plants design, operation and costs
- **- 1 pts** Missing how the environmental aspects impact on nuclear power plant design, operation and cost
- **- 0.5 pts** Missing a list of the references used
- **- 0.5 pts** Missing the percentage for coal
- **- 2 pts** It's not clearly stated the impact of all environmental considerations mentioned for all four sources of power plants
- **- 0.5 pts** Did not separate the percentage of fossil fuel energy into natural gas and coal
- **- 2.5 pts** Did not list at least 3 (three) environmental aspects and their impact plant design, operation and cost for nuclear and hydroelectric power plants

- 0.5 pts Listed only 2 environmental considerations for natural gas power plants

- 0.5 pts Listed only 2 environmental considerations for coal power plants

- 1.5 pts Listed only 1 environmental aspect for hydroelectric power plant

- 1.5 pts It's not clearly stated how those environmental aspects impact plant design, operation and cost for coal and natural gas power plants

- 1.5 pts It's not clearly stated how those environmental aspects impact plant design, operation and cost for nuclear and hydroelectric power plants